



Department of Education

**STEM Education**  
**National Schools of Excellence**  
**STEM Physics Teachers Guide**  
**Grade 11**  
**Term 2**

| <b><u>DOMAIN</u></b> | <b><u>Phylum</u></b> |
|----------------------|----------------------|
| <b>SCIENCE</b>       | <b>Physics</b>       |
| <b>Technology</b>    |                      |
| <b>Engineering</b>   |                      |
| <b>Mathematics</b>   |                      |

Issued free to schools by the Department of Education

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## **Acknowledgements**

This STEM Physics Teacher Guide for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

This document was developed with support from the General Education Services Division of the Department of Education.

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## SECRETARY'S MESSAGE

This Physics Teacher Guide is to be used exclusively by teachers of Physics to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Physics Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Physics before undertaking the advanced concepts introduced in this syllabus.

The Teacher Guide conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Physics as a fundamental field of science.

This STEM Physics Teacher Guide for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Physics.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Physics is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Physics to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this Teacher Guide as the official curriculum for Physics to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD  
Secretary for Education

# INTRODUCTION

Physics is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Physics help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Physics is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Physics.

The Grade 11 Physics Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Physics is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Teacher Guide are sequenced to guide teachers of Physics through the process of imparting basic chemistry knowledge and skills to students.

| THEMA                   | MODULE                      | TOPICS                                                                                      |
|-------------------------|-----------------------------|---------------------------------------------------------------------------------------------|
| BIODIVERSITY            | Ethnobotany                 | 1. Medicinal/Herbal Plants<br>2. Pharmacology of Traditional Plants<br>3. Therapeutic Drugs |
| BIOTECHNOLOGY           | Principles of Biotechnology | 1. Polymerase Chain Reaction<br>2. Dye Chemistry in Gene Sequencing<br>3. Proteins          |
| HEALTH AND MEDICINE     | Biochemistry in Medicine    | 1. Drug Design<br>2. Pharmacology<br>3. Significance of Diagnostic Laboratory Tests         |
| EARTH SCIENCE           |                             |                                                                                             |
| WASTE MANAGEMENT        |                             |                                                                                             |
| INFRASTRUCTURE          |                             |                                                                                             |
| ENERGY                  |                             |                                                                                             |
| BATTERY TECHNOLOGY      |                             |                                                                                             |
| INFORMATION SYSTEM      |                             |                                                                                             |
| COMPUTER SCIENCE        |                             |                                                                                             |
| ARTIFICIAL INTELLIGENCE |                             |                                                                                             |
| ENTERPRENEURSHIP        |                             |                                                                                             |

# RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Teacher Guide is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Physics Bridging Teacher Guide students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Physics as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Teacher Guide. This Teacher Guide will be introduced in the course of time.

# **STEM Domain**

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

## **Science**

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

## **Technology**

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

## **Engineering**

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

## **Mathematics**

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

## STEM PHYSICS CONTENT OVERVIEW

**Table 1.1: Thema Sequence and Duration**

| Grade 11 Term 2: STEM-proper Course     |                                                               |                                                                                                                                                                                                        |
|-----------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BIODIVERSITY</b><br>(3 Weeks)        | 11.1 Measuring Biodiversity and Waves                         | <ol style="list-style-type: none"> <li>1. Electromagnetic Waves in Vegetation Monitoring</li> <li>2. Electromagnetic Waves in Forest Mapping</li> <li>3. Sound Waves in Identifying Animals</li> </ol> |
| <b>BIOTECHNOLOGY</b><br>(3 Weeks)       | 11.1 Measuring Biomolecules through Electricity and Magnetism | <ol style="list-style-type: none"> <li>1. DNA-Transistor</li> <li>2. Magnetic Resonance</li> <li>3. Force Spectroscopy</li> </ol>                                                                      |
| <b>HEALTH AND MEDICINE</b><br>(3 Weeks) | 11.1 Medical Physics                                          | <ol style="list-style-type: none"> <li>1. Ultrasound</li> <li>2. X-ray</li> <li>3. CT-Scan</li> </ol>                                                                                                  |

**Table 1.2 Sequencing & Benchmarking STEM Physics Content Standards**

| Setting STEM Physics Content Standards for Progressive Learning & Assessment<br>National STEM Assessment |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Domain                                                                                                   | Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Module Content Standard                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                       |
| <b>THEMA</b>                                                                                             | <b>BIODIVERSITY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 11.1: Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world.                                                                                                                                                                                                                                                 |                                                                                                                                       |
| <b>Module 11.1</b>                                                                                       | <b>Measuring Biodiversity and Waves</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                       |
| Topics                                                                                                   | Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Module Benchmark                                                                                                                                                                                                                                                                                                                                                                   | Assessment Tasks                                                                                                                      |
| 1. Electromagnetic Waves in Vegetation Monitoring                                                        | <b>11.1.1a:</b> Define Waves and Electromagnetic Waves<br><b>11.1.1b</b> Differentiate between Absorption, Transmission and Reflectance behavior of Wave<br><b>11.1.1c:</b> Identify the Electromagnetic Spectrum<br><b>11.1.1d:</b> Define Radiation<br><b>11.1.1e:</b> Classify the different types of Radiation<br><b>11.1.1f:</b> Describe how Radiation Interacts with Matter<br><b>11.1.1g:</b> Explain how satellite sensors works.<br><b>11.1.1h</b> Define Spectral signature of objects<br><b>11.1.1i</b> Discuss how RGB images work<br><b>11.1.1j</b> Explain how health monitoring of vegetation works | Student's ability to master knowledge and skills of; <ul style="list-style-type: none"> <li>• Waves</li> <li>• Electromagnetic Wave Principles</li> <li>• Radiation Principles</li> <li>• Satellite Imaging</li> <li>• Vegetation Mapping</li> <li>• Forest Mapping</li> <li>• Sound Waves</li> <li>• Sound Conversion Process</li> <li>• Animal Identification Process</li> </ul> | <ul style="list-style-type: none"> <li>• Assignment</li> <li>• Module Test</li> <li>• Lab Work</li> <li>• End of Term Exam</li> </ul> |
| 2. Electromagnetic Waves in Forest Mapping                                                               | <b>11.1.2a:</b> Understand that Forest mapping process is similar to Vegetation Monitoring<br><b>11.1.2b:</b> Explain how Forest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                       |



|                                       |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|                                       | Mapping process works                                                                                                                                                                                                                                          |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| 3. Sound Waves in Identifying Animals | <b>11.1.3a:</b> Define Sound Waves<br><b>11.1.3b:</b> Understand Sound Production Mechanism<br><b>11.1.3c:</b> Explain the process of Sound Conversion: Analog to Digital<br><b>11.1.3d:</b> Discuss and describe how animals sound are captured and recorded. |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| THEMA                                 | BIOTEHCNOLOGY                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| Module 11.1                           | Measuring Biomolecules Through Electricity and Magnetism                                                                                                                                                                                                       | Module Content Standard                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
|                                       |                                                                                                                                                                                                                                                                | <b>11.2a:</b> Discuss and describe the process of measuring biomolecules<br><b>11.2b:</b> Describe and analyze DNA-Transistor, Magnetic Resonance and Force Spectroscopy<br><b>11.2c:</b> Define Electricity and Magnetism |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| Topics                                | Performance Indicators                                                                                                                                                                                                                                         |                                                                                                                                                                                                                            | Module Benchmark                                                                                                                                                                                                                                                                                                                           | Assessment Tasks                                                                                                         |
| 1. DNA-Transistor                     | <b>11.1.1a:</b> Define and describe the Electric Principles forming the DNA-transistor<br><b>11.1.1b:</b> Define DNA-Transistor<br><b>11.1.1c:</b> Understand the process of DNA Sequencing using DNA-Transistor                                               |                                                                                                                                                                                                                            | Student's ability to master knowledge and skills of;<br><ul style="list-style-type: none"><li>Principles of Electricity</li><li>Principles of Magnetism</li><li>DNA-Transistor</li><li>Magnetic Resonance</li><li>Force Spectroscopy</li><li>Measurement</li><li>Biomolecule Types</li><li>DNA Sequencing</li><li>Force Analysis</li></ul> | <ul style="list-style-type: none"><li>Assignment</li><li>Module Test</li><li>Lab Work</li><li>End of Term Exam</li></ul> |
| 2. Magnetic Resonance                 | <b>11.1.2a:</b> Describe and analyse Magnetic Resonance<br><b>11.1.2b</b> Define Magnetism Principles used in Magnetic Resonance<br><b>11.1.2c</b> Recognize importance of Magnetic Resonance in Diagnosis and Treatment                                       |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| 3. Force Spectroscopy                 | <b>11.1.3a:</b> Define Mechanical Forces<br><b>11.1.3b:</b> Define Binding Forces<br><b>11.1.3c:</b> Define Force Spectroscopy and describe its process<br><b>11.1.3d:</b> Apply Force Spectroscopy Techniques to measure biomolecules                         |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| THEMA                                 | HEALTH AND MEDICINE                                                                                                                                                                                                                                            |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| Module 11.1                           | Medical Physics                                                                                                                                                                                                                                                | Module Content Standard                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
|                                       |                                                                                                                                                                                                                                                                | <b>11.1.a:</b> Explain medical diagnosis using ultrasound<br><b>11.1.b:</b> Explain medical diagnosis and analysis using X-ray<br><b>11.1.c:</b> Explain medical diagnosis and analysis using Computerized Tomography Scan |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                          |
| Topics                                | Performance Indicators                                                                                                                                                                                                                                         |                                                                                                                                                                                                                            | Module Benchmark                                                                                                                                                                                                                                                                                                                           | Assessment Tasks                                                                                                         |

|                                 |                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                    |                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1. Ultrasound                   | <b>11.1.1a:</b> Define sound and differentiate ultrasound<br><b>11.1.1b:</b> Explain ultrasound in the Echolocation Process<br><b>11.1.1c:</b> Apply Echolocation to identify fetus in the womb            | Student's ability to master knowledge and skills of; <ul style="list-style-type: none"> <li>Principles of Medical Diagnosis and Treatment</li> <li>Application of Sound Waves</li> <li>Application of X-Ray waves</li> <li>Application of Computerized Tomography Scan.</li> </ul> | <ul style="list-style-type: none"> <li>Assignment</li> <li>Module Test</li> <li>Lab Work</li> <li>End of Term Exam</li> </ul> |
| 2. X-Ray                        | <b>11.1.2a:</b> Define and Describe X-ray Waves<br><b>11.1.2b:</b> Discuss How X-ray is used in Medical Diagnosis<br><b>11.3.2c:</b> State some of the application of X-ray techniques                     |                                                                                                                                                                                                                                                                                    |                                                                                                                               |
| 3. Computerized Tomography Scan | <b>11.1.3a:</b> Define Computerized Tomography<br><b>11.1.3b:</b> Discuss how Computerized Tomography Works<br><b>11.1.3c:</b> Explain the process of medical diagnosis using Computerized Tomography Scan |                                                                                                                                                                                                                                                                                    |                                                                                                                               |

**Table 1.3 Mapping Content Progressions by Modules**

| Thema/Module                                                                           | Broad Content Standard                                                                                                                                                                                                     | 11.1 | 11.1 | 11.1 |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|
| <b>Biodiversity/</b><br>11.1 Measuring Biodiversity and Waves                          | <b>11.1:</b> Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world.                                                                                  | ✓    |      |      |
| <b>Biotechnology/</b><br>11.1 Measuring Biomolecules through Electricity and Magnetism | <b>11.1a:</b> Discuss and describe the process of measuring biomolecules<br><b>11.1b:</b> Describe and analyze DNA-Transistor, Magnetic Resonance and Force Spectroscopy<br><b>11.1c:</b> Define Electricity and Magnetism |      | ✓    |      |
| <b>Health and Medicine/</b><br>11.1 Medical Physics                                    | <b>11.1a:</b> Explain medical diagnosis using ultrasound<br><b>11.1b:</b> Explain medical diagnosis and analysis using X-ray<br><b>11.1c:</b> Explain medical diagnosis and analysis using Computerized Tomography Scan    |      |      | ✓    |

# **THEMA BIODIVERSITY**

## MODULE 11.1: MEASURING BIODIVERSITY AND WAVES

### **3weeks**

In this module we will see how Physics is used in the aid of measuring Biodiversity. Physics is the study of the Laws of Nature and those Laws are applicable to Biodiversity in many ways. Measuring Biodiversity will help Biologists, Environmental scientists and other dedicated scientists who want those measurements to make better informed decision. In addition, it will broaden their knowledge in understanding the distribution of Biodiversity across the planet. This module will enable the student to grasp the real-life skill of measuring and interpreting data based on physical laws.

### **Broad Content Standard**

**11.1** Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world.

### **Content Benchmark**

Student's ability to master knowledge and skills of;

- Waves
- Electromagnetic Wave Principles
- Radiation Principles
- Satellite Imaging
- Vegetation Mapping
- Forest Mapping
- Sound Waves
- Sound Conversion Process
- Animal Identification Process

# Topic 1: Electromagnetic Waves in Vegetation Monitoring

## Performance Indicators

Students are expected to:

11.1.1a: Define Waves and Electromagnetic Waves

11.1.1b Differentiate between Absorption, Transmission and Reflectance behavior of Wave

11.1.1c: Identify the Electromagnetic Spectrum

11.1.1d: Define Radiation

11.1.1e: Classify the different types of Radiation

11.1.1f: Describe how Radiation Interacts with Matter

11.1.1g: Explain how satellite sensors works.

11.1.1h: Define Spectral signature of objects

11.1.1i: Discuss how RGB images work

11.1.1j: Explain how health monitoring of vegetation works

## Activity 1: Classroom Activity

### *Title: Electromagnetic Waves*

- Discuss in pairs to;
  - Define Waves and Electromagnetic Waves
  - Identify Electromagnetic wave as a Type of wave and describe the Electromagnetic Wave in terms of Energy, Frequency, Velocity and Wavelength
  - Put some practical problems on the board for students to copy and do one as an example and assign others for students to do as part of their homework.

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 2: Classroom Activity

### *Title: Differentiate between Absorption, Transmission and Reflectance behavior of Wave*

- Do activities on how a travelling wave behaves as it comes into contact with objects;
  - Light wave Absorption
  - Light Wave Transmission
  - Light Wave Reflection
- Describe how travelling waves also are scattered, refracted and diffracted.

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 3: Classroom Activity

**Title:** *Identify the Electromagnetic Spectrum*

- Set activities for students to solve on;
  - Identifying the different Electromagnetic Waves that make up the spectrum based on these properties:
    - Frequency
    - Wavelength
    - Energy

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 4: Classroom Activity

**Title:** *Radiation*

- Set activities for students to explain the concepts of:
  - Radiation
  - High and Low Radiation
  - Ionizing and Non-ionizing Radiation

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 5: Classroom Activity

**Title:** *Classify the different types of Radiation*

- Discuss and Describe the different types of Radiation: (Work in groups)
  - Alpha rays
  - Beta rays
  - Neutrons
  - Gamma rays
  - Electromagnetic Radiation
- Discuss and Describe the following forms of Radiation: (Work in groups)
  - Solar Radiation
  - Visible Light Radiation
  - Thermal Radiation
  - Infrared Radiation

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 6: Classroom Activity

### *Title: Radiation Interaction*

- Get students to work in groups to define, identify and differentiate between Radiation Interactions with matter and Waves interactions with matter to understand Wave behavior:
  - Absorption Behavior
  - Reflection Behavior
  - Transmission Behavior
  - Radiation Behavior of matter

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 7: Classroom Activity

### *Title: How Satellite Sensors Work*

- Set activity to define:
  - Remote Sensing
  - Satellite Sensors
- Discuss in pairs to explain satellite sensors with bands/channels within following domains:
  - Visible Spectrum
  - Near Infrared Spectrum
  - Middle Infrared spectrum

- Differentiate between Active and Passive Remote Sensors
- Set Activities for them to see the type of data collected by satellite sensors

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 8: Classroom Activity

### *Title: Spectral Signatures of Object*

- Define Spectral Signature
- Define Reflectance Spectra
- Get students to discuss the spectral signature: Visible Spectrum, Near infrared and Middle infrared of the following objects:
  - Grass
  - Sand
  - Vegetative Canopy

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 9: Classroom Activity

### *Title: RGB Image*

- Define RGB Image
- Describe RGB Techniques
- Explain how RGB image can be used to extract information

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 10: Classroom Activity

### *Title: Health Monitoring of Vegetation*

- Define Remote Health monitoring of Vegetation
- Discuss how RGB aids to give information about vegetation cover
- Describe how Visible, Near infrared and Middle infrared Reflectance Spectra reveals health information of the vegetation.

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)



## Topic 2: Electromagnetic Waves in Forest Mapping

### Performance Indicators

Students are expected to;

11.1.2a: Define Forest Mapping

11.1.2b: Identify the electromagnetic waves used in Forest Mapping

11.1.2c: Differentiate between Forest Mapping and Vegetation Monitoring

11.1.2d: Explain how Forest Mapping process works

### Activity 1: Classroom Activity

#### *Title: Defining Forest Mapping*

- Discuss in pairs to;
  - Define Forest Mapping
  - Compare and Contrast Forest Mapping to Vegetation Monitoring

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

#### *Title: How Forest Mapping Works*

- Do activities on Spectral Signature as an indicator for Forest Mapping

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Topic 3: Sound Waves in Identifying Animals

### Performance Indicators

Students are expected to:

11.1.3a: Define Sound Waves

11.1.3b: Understand Sound Production Mechanism

11.1.3c: Explain the process of Sound Conversion: Analog to Digital

11.1.3d: Discuss and describe how animals sound are captured and recorded.

### Activity 1: Classroom Activity

#### *Title: Defining Sound Waves*

- Define Sound Waves
- Explain Sound Behavior

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

#### *Title: Sound Production Mechanism*

- Discuss in pairs to;
  - Definition of Vibration, Oscillation and Frequency
  - Explain how sound is produced

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 3: Classroom Activity

#### *Title: Sound Conversion – Analogue to Digital*

- Explain the concept of Analog to Digital conversion of electronic signals
- Explain how sound is converted to digital signal

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 4: Classroom Activity

#### *Title: How Animals are captured and recorded*

- Discuss in pairs to;
  - Explain the process of animal sound recording, storing and identification process

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

# SAMPLE ASSESSMENTS for BIODIVERSITY

## Assessment Methods (Teachers can decide)

- Assignment
- Lab Work
- Research Case Studies
- End of Term Exam

## Assessment Tasks

| Module                                     | Content Standard                                                                                                             | Topic                                             | Assessment Tasks |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------|
| <b>1. Measuring Biodiversity and Waves</b> | Discuss and analyze the importance of Biodiversity's measurements and its application in solving problems in the real world. | 1. Electromagnetic Waves in Vegetation Monitoring | Assignment       |
|                                            |                                                                                                                              | 2. Electromagnetic Waves in Forest Mapping        |                  |
|                                            |                                                                                                                              | 3. Sound Waves in Identifying Animals             |                  |

## Sample Marking Guide

| MODULE<br>Topics                                           | Assessment<br>Task | Assessment Criteria                                                                | Total<br>Marks |
|------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------|----------------|
|                                                            |                    | Students will be tested on the ability to;                                         |                |
| 1. Electromagnetic<br>Waves in<br>Vegetation<br>Monitoring | Assignment         | Concept of Electromagnetic Waves and<br>Process of Vegetation’s Health Monitoring  | 10             |
| Electromagnetic<br>Waves in Forest<br>Mapping              |                    | Process of Forest Mapping                                                          | 10             |
| Sound Waves in<br>Identifying Animals                      |                    | Concept of Sound, Animal Sound<br>Conversion Process and Identification<br>Process | 10             |
| TOTAL Mark                                                 |                    | 30                                                                                 |                |

# **THEMA**

# **BIOTECHNOLOGY**

# MODULE 11.1: MEASURING BIOMOLECULES THROUGH ELECTRICITY AND MAGNETISM

## **3weeks**

Biomolecules are critical elements in all living things. They come in different sizes at a microscopic and nanoscopic levels at the cellular level. By analyzing their physical size, this could provide vital insights into their structure and function. Biomolecules are classed into four main groups known as Carbohydrates, Lipids, Nucleic-acid and Proteins. In this module, we'll take a look at the different physical techniques used in analyzing Biomolecules. This module will be more basic in explaining concepts in a systematic manner without emphasizing on the calculations.

### **Broad Content Standard**

**11.1** Discuss and analyze the importance of Biotechnological devices and its application in understanding Biomolecules.

### **Content Benchmark**

Student's ability to master knowledge and skills of;

- Principles of Electricity
- Principles of Magnetism
- DNA-Transistor
- Magnetic Resonance
- Force Spectroscopy
- DNA Sequencing
- Force Analysis

## Topic 1: DNA-Transistor

### Performance Indicators

Students are expected to;

11.1.1a: Define DNA-Transistor

11.1.1b: Define and describe the Electric Principles applied in the DNA-transistor

11.1.1c: Understand the process of DNA Sequencing using DNA-Transistor

### Activity 1: Classroom Activity

**Title:** *Electric Principles Forming DNA Transistor*

- Discuss in pairs to;
- Explain the concept of Transistor
- Define Voltage
- Define Electric Current
- Define Electric Charge

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

**Title:** *DNA Transistor*

- Explain the concept of DNA-Transistor

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 3: Classroom Activity

**Title:** *Process of DNA Sequencing using DNA-Transistor*

- Explain the process of DNA Sequencing using DNA-Transistor

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Topic 2: Magnetic Resonance Spectroscopy

### Performance Indicators

Students are expected to;

11.1.2a: Describe and analyse Magnetic Resonance

11.1.2b: Define Magnetism Principles used in Magnetic Resonance

11.1.2c: Recognize importance of Magnetic Resonance in Diagnosis and Treatment

### Activity 1: Classroom Activity

#### *Title: Magnetic Resonance*

- Discuss in pairs to Describe the concept of Magnetic Resonance Spectroscopy

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

#### *Title: Magnetic Principles in Magnetic Resonance*

- Describe the concept of Magnetism
- Define Magnetic Field
- Describe Magnetic Field Lines

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 3: Classroom Activity

#### *Title: Magnetic Resonance in Diagnosis and Treatment*

- Describe the process of Magnetic resonance spectroscopy in identifying biomolecules

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Topic 3: Force Spectroscopy

### Performance Indicators

Students are expected to;

11.1.3a: Define Mechanical Forces

11.1.3b: Define Binding Forces

11.1.3c: Define Force Spectroscopy and describe its process

11.1.3d: Apply Force Spectroscopy Techniques to measure biomolecules

### Activity 1: Classroom Activity

#### *Title: Force Spectroscopy and its Process*

- Discuss in pairs and explain the concept of Force Spectroscopy and its Process Spectroscopy

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

#### *Title: Force Spectroscopy Techniques to Measure Biomolecules*

- Discuss in pairs to explain the process of Force Spectroscopy
- **Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)



# SAMPLE ASSESSMENTS for BIOTECHNOLOGY

## Assessment Methods (Teachers can decide)

- Assignment
- Lab Work
- Research Case Studies
- End of Term Exam

## Assessment Tasks

| Module                                                          | Broad Content Standard                                                                                                                         | Topic                              | Assessment Tasks |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------|
| <b>Measuring Biomolecules Through Electricity and Magnetism</b> | <b>11.1a:</b> Describe and analyze DNA-Transistor, Magnetic Resonance and Force Spectroscopy<br><b>11.1b:</b> Define Electricity and Magnetism | 1. DNA-Transistor                  | Assignment 1     |
|                                                                 |                                                                                                                                                | 2. Magnetic Resonance Spectroscopy |                  |
|                                                                 |                                                                                                                                                | 3. Force Spectroscopy              |                  |

## Sample Marking Guide

| MODULE<br>Topics                   | Assessment<br>Task | Assessment Criteria                                                                | Total<br>Marks |
|------------------------------------|--------------------|------------------------------------------------------------------------------------|----------------|
|                                    |                    | Students will be tested on the ability<br>to;                                      |                |
| DNA-Transistor                     | Assignment         | Concept of DNA Transistor and<br>Process of DNA Sequencing using<br>DNA Transistor | 10             |
| Magnetic Resonance<br>Spectroscopy |                    | Concept and Process of Magnetic<br>Resonance Spectroscopy                          | 10             |
| Force Spectroscopy                 |                    | Concept of Force Spectroscopy and<br>Application Process of Force<br>Spectroscopy  | 10             |
| TOTAL Mark                         |                    |                                                                                    | 30             |

# **THEMA HEALTH AND MEDICINE**

## MODULE 11.1: MEDICAL PHYSICS

### 3 weeks

Medical Physics is a study that basically utilizes the concept of Radiation in medical analysis to help diagnose patients. One of its main functions is to analyze the internal structures of the human body through imaging techniques. Medical physics is also used as treatment, prevention and diagnostic measures for diseases. It is very critical in hospitals and healthcare industries. By understanding medical physics, health workers may have a very handy tool in correctly analyzing sick patients.

### **Broad Content Standard**

**11.1** Discuss and analyze the importance of Medical Physics and its application in medical diagnosis and treatment.

### **Content Benchmark**

Student's ability to master knowledge and skills of;

- Principles of Medical Diagnosis and Treatment
- Application of Sound Waves
- Application of X-Ray waves
- Application of Computerized Tomography Scan

# Topic 1: Ultrasound

## Performance Indicators

Students are expected to;

- 11.1.1a: Define sound and differentiate ultrasound
- 11.1.1b: Explain ultrasound in the Echolocation Process
- 11.1.1c: Apply Echolocation to identify fetus in the womb

## Activity 1: Classroom Activity

**Title:** *Sound and Ultrasound*

- Explain the concept of Ultrasound with respect to Sound

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 2: Classroom Activity

**Title:** *Ultrasound in Echolocation Process*

- Discuss in pairs to
  - Define Echolocation
  - Describe why ultrasound is used in Echolocation
  - Describe how ultrasound is used in Echolocation

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Activity 3: Classroom Activity

**Title:** *Echolocation to Identify Fetus*

- Define Obstetric
- Describe the Process of Obstetric

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Topic 2: X-Ray

### Performance Indicators

Students are expected to;

11.1.2a: Define and Describe X-ray Waves

11.1.2b: Discuss How X-ray is used in Medical Diagnosis

11.1.2c: State some of the application of X-ray techniques

### Activity 1: Classroom Activity

#### *Title: X-ray Waves*

- Define X-ray
- Identify X-ray in the Electromagnetic Spectrum
- Describe the properties of X-rays

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

#### *Title: X-ray in Medical Diagnosis*

- Discuss in pairs:
  - Explain the X-ray production
  - Describe how X-ray is applied in Medical Diagnosis

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 3: Classroom Activity

#### *Title: Application of X-ray Techniques*

- List and Explain some of the application of X-ray techniques

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

## Topic 3: Computerized Tomography Scan

### Performance Indicators

Students are expected to;

11.1.3a: Define Computerized Tomography

11.1.3b: Discuss how Computerized Tomography Works

11.1.3c: Explain the process of medical diagnosis using Computerized Tomography Scan

11.3.2c: State some of the application of CT Techniques

### Activity 1: Classroom Activity

#### ***Title: CT-Scan***

- Define Computerized Tomography Scan
- Differentiate between X-Ray and Computerized Tomography Scan

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 2: Classroom Activity

#### ***Title: How CT-Scan Works***

- Discuss in pairs to explain the process of Computerized Tomography Scan

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

### Activity 3: Classroom Activity

#### ***Title: Process of Medical Diagnosis using CT-Scan***

- Discuss in pairs to explain how doctors use CT-Scan in their Medical Diagnosis

**Resources:** White board markers, chalks, dusters, meter rulers, resource books and handouts.

(Refer to Physics TRB)

# SAMPLE ASSESSMENTS for HEALTH AND MEDICINE

## Assessment Methods (Teachers can decide)

- Assignment
- Lab Work
- Research Case Studies
- End of Term Exam

## Assessment Tasks

| Module                 | Content Standard                                                                                             | Topic                           | Assessment Tasks |
|------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------|------------------|
| <b>Medical Physics</b> | <b>11.1:</b> Explain medical diagnosis and analysis using Ultrasound, X-ray and Computerized Tomography Scan | 1. Ultrasound                   | Assignment 1     |
|                        |                                                                                                              | 2. X-ray                        |                  |
|                        |                                                                                                              | 3. Computerized Tomography Scan |                  |

## Sample Marking Guide

| MODULE Topics                   | Assessment Task | Assessment Criteria                                                   | Total Marks |
|---------------------------------|-----------------|-----------------------------------------------------------------------|-------------|
|                                 |                 | Students will be tested on the ability to;                            |             |
| 1. Ultrasound                   | Assignment      | Concept of Ultrasound and Process of Echolocation in Fetus Imaging    | 10          |
| 2. X-Ray                        |                 | Concept of X-Ray and Process of X-ray Analysis                        | 10          |
| 3. Computerized Tomography Scan |                 | Concept of Computerized Tomography Scan and Process of X-ray Analysis | 10          |
|                                 | TOTAL Mark      |                                                                       | 30          |

# PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

## Assessment Planning

| Assessment Domains                            | Modules          |                |                  |                | Total Marks | % Weight of Modules |
|-----------------------------------------------|------------------|----------------|------------------|----------------|-------------|---------------------|
|                                               | 1                |                | 2                | 3              |             |                     |
| Theory Knowledge & skills – <b>Test</b>       | Test (50)        |                | Test (50)        | Test (50)      | /150        | <b>20</b>           |
| Application of knowledge – <b>Assignments</b> | Assign.1 (25)    | Assign. 2 (25) | Assign.3 (25)    | Assign.4 (25)  | /100        | <b>30</b>           |
| End of Term <b>Exam</b>                       | 35% (17.5 marks) |                | 35% (17.5 marks) | 20% (15 marks) | /50         | <b>50</b>           |
| Total marks for term one                      |                  |                |                  |                |             | <b>100 marks</b>    |

Outlined in the table above is an overall assessment per module for the 3 modules of the 2<sup>nd</sup> term of Grade 11. It covers the key domains of the four modules. This is only a guide and the teacher is given the autonomy of setting whatever assessment task that is necessary for a given topic, however the teacher is required to comply with the assessment guidelines set above for uniformity throughout the country.

## Assessment Recording

| Percentage Range | Achievement Level               |
|------------------|---------------------------------|
| 85 -100          | Very High Achievement (VHA)     |
| 65-84            | High Achievement (HA)           |
| 50-64            | Satisfactory Achievement (SA)   |
| 25-49            | Low Achievement (LA)            |
| 0-24             | Below Minimum Achievement (BMA) |

| # | Names    | Assessment Task |    |    |             |    |    |    |            | Total Score (/300) | Percentage Weight (%) |         |            |        | Achievement Level [VHA, HA, SA,LA] |
|---|----------|-----------------|----|----|-------------|----|----|----|------------|--------------------|-----------------------|---------|------------|--------|------------------------------------|
|   |          | Tests           |    |    | Assignments |    |    |    | Exam (/50) |                    | T (20%)               | A (30%) | Exam (50%) | Ttl CA |                                    |
|   |          |                 |    |    |             |    |    |    |            |                    |                       |         |            |        |                                    |
| 1 | 2        | 3               | 1  | 2  | 3           | 4  |    |    |            |                    |                       |         |            |        |                                    |
| 1 | Bill Joe | 45              | 47 | 43 | 20          | 21 | 22 | 23 | 45         | 246                | 18                    | 25.8    | 45         | 88.8   | VHA                                |
| 2 | Cara Den | 20              | 25 | 27 | 15          | 16 | 17 | 18 | 30         | 168                | 9.6                   | 19.8    | 30         | 59.4   | SA                                 |
| 3 | Lea May  | 9               | 15 | 20 | 5           | 6  | 7  | 8  | 25         | 115                | 5.87                  | 7.7     | 25         | 38.7   | LA                                 |



## Assessment Reporting

| Performance Standards For <b>Assignment 1:</b><br>Module 1: |                                                                                     |                                                                                            |                                                                             |                |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------|
| Performance Criteria                                        | Accurately answers All the Questions (15 marks)                                     | Precisely explain with examples (5 marks)                                                  | Correctly explain how the number system work in computers (5 marks)         | 25 marks       |
| <b>Very High Achievement</b>                                | Accurately interprets the questions and write the correct answers (13-15)           | Accurately interpret and precisely answers and write the correct examples (5)              | Accurately explain how the number system works in computer (5)              | <b>(21-25)</b> |
| <b>High Achievement</b>                                     | Partially interprets the questions and write the correct answers (10-12)            | Partially interpret and precisely answers and write the correct examples (4)               | Partially explain how the number system works in computer (4)               | <b>(16-20)</b> |
| <b>Satisfactory Achievement</b>                             | Fairly interprets the questions and write the correct answers (7-9)                 | Fairly interpret and precisely answers and write the correct examples (3)                  | Impartially explain how the number system works in computer (3)             | <b>(11-15)</b> |
| <b>Low Achievement</b>                                      | Poorly interprets the questions and write the correct answers (4-6)                 | Poorly interpret and precisely answers and write the correct examples (2)                  | Poorly explain how the number system works in computer (2)                  | <b>(6-10)</b>  |
| <b>Below minimum Achievement</b>                            | Needs improvement in interpreting the questions and write the correct answers (0-3) | Needs improvement in interpreting and precisely answers and write the correct examples (1) | Needs improvement in explaining how the number system works in computer (1) | <b>(0-5)</b>   |

| <b>Performance Standards For Test:<br/>Module 1:</b> |                                                        |                                                          |                                                                                            |                 |
|------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------|
| <b>Performance Criteria</b>                          | <b>Correctly answers all long questions (25 marks)</b> | <b>Correctly answers all multiple choice (10 marks)</b>  | <b>Correctly writes the correct conversion of binary to data and vice versa (15 marks)</b> | <b>50 marks</b> |
| <b>Very High Achievement</b>                         | Accurately answers all long questions (21-25)          | Accurately answers all multiple choice (9-10)            | Accurately writes the correct conversion of binary to data and vice versa (13-15)          | <b>(41-50)</b>  |
| <b>High Achievement</b>                              | Partially answers all long questions (16-20)           | Partially answers all multiple choice (7-8)              | Partially writes the correct conversion of binary to data and vice versa (10-12)           | <b>(31-40)</b>  |
| <b>Satisfactory Achievement</b>                      | Fairly answers all long questions (11-15)              | Fairly answers all multiple choice (5-6)                 | Fairly writes the correct conversion of binary to data and vice versa (7-9)                | <b>(21-30)</b>  |
| <b>Low Achievement</b>                               | Poorly answers all long questions (6-10)               | Poorly answers all multiple choice (3-4)                 | Poorly writes the correct conversion of binary to data and vice versa (4-6)                | <b>(11-20)</b>  |
| <b>Below minimum Achievement</b>                     | Needs improvement in answering long answer questions   | Needs improvement in answering all multiple choice (0-2) | Needs improvement in writing the correct conversion of binary to data and vice versa (0-3) | <b>(1-10)</b>   |

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# GLOSSARY

| Words                     | Meaning                                                                                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| absorption band           | wavelength interval within which electromagnetic radiation is absorbed by the atmosphere or by other substances                                                                                                   |
| active remote sensing     | remote sensing methods that provide their own source of electromagnetic radiation                                                                                                                                 |
| atmospheric window        | wavelength interval within which the atmosphere readily transmits electromagnetic radiation                                                                                                                       |
| band                      | a wavelength interval in the electromagnetic spectrum. For example, in Landsat images the bands designate specific wavelength intervals at which images are acquired                                              |
| detector                  | component of a remote sensing system that converts electromagnetic radiation into a recorded signal                                                                                                               |
| echoes                    | Reflections or repetitions of sound waves.                                                                                                                                                                        |
| electromagnetic radiation | energy propagated in the form of and advancing interaction between electric and magnetic fields. All electromagnetic radiation moves at the speed of light                                                        |
| emission                  | process by which a body radiates electromagnetic energy. Emission is determined by kinetic temperature and emissivity                                                                                             |
| frequency                 | The number of times a vibrating object oscillates (moves back and forth) in one second. Fast movements produce high frequency sound (high pitch/tone), but slow movements mean the frequency (pitch/tone) is low. |
| ground receiving station  | facility that records data transmitted by a satellite, such as Landsat                                                                                                                                            |
| image                     | pictorial representation of a scene recorded by a remote sensing system. Although image is a general term, it is commonly restricted to representations acquired by non-photographic methods                      |
| landsat                   | a series of unnamed earth-orbiting NASA satellites that acquire multispectral images in various visible and Infrared bands                                                                                        |
| laser                     | light artificially stimulated electromagnetic radiation: a beam of coherent radiation with a single wavelength                                                                                                    |
| lidar                     | light intensity detection and ranging, which uses lasers to stimulate fluorescence in various compounds and to measure distances to                                                                               |

|                        |                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | reflecting surfaces                                                                                                                                                                                                                                                                                                                                                                     |
| panchromatic film      | black and white film that is sensitive to all visible wavelengths                                                                                                                                                                                                                                                                                                                       |
| passive remote sensing | remote sensing of energy naturally reflected or radiated from the terrain                                                                                                                                                                                                                                                                                                               |
| photodetector          | device for measuring energy in the visible-light band                                                                                                                                                                                                                                                                                                                                   |
| photon                 | minimum discrete quantity of radiant energy                                                                                                                                                                                                                                                                                                                                             |
| piezoelectricity       | The generation of electricity or of electric polarity in dielectric crystals subjected to mechanical stress, or the generation of stress in such crystals subjected to an applied voltage.                                                                                                                                                                                              |
| pixel                  | contraction of picture element                                                                                                                                                                                                                                                                                                                                                          |
| pulse                  | short burst of electromagnetic radiation transmitted by a radar antenna                                                                                                                                                                                                                                                                                                                 |
| radar                  | Acronym for radio detection and ranging. Radar is an active form of remote sensing that operates in the microwave and radio wavelength regions                                                                                                                                                                                                                                          |
| raster format          | a means of representing spatial data in the form of a grid of DN, each line of which can be used to modulate the lines of a video raster                                                                                                                                                                                                                                                |
| semiconductor          | A substance, usually a solid chemical element or compound that can conduct electricity under some conditions but not others, making it a good medium for the control of electrical current. Its conductance varies depending on the current or voltage applied to a control electrode, or on the intensity of irradiation by infrared (IR), visible light, ultraviolet (UV), or X rays. |